

Good evening! That is a beautifully intuitive way to think about it. Heating a metal sheet gives us a perfect physical canvas for differential geometry.

To prove that the gradient of your temperature function f transforms as a **covariant vector** (or a dual vector/one-form), we need to look at how its components change when you change your coordinate system.

The punchline of the proof is this: **a covariant vector transforms using the partial derivatives of the *old* coordinates with respect to the *new* ones**. This is a direct consequence of the multivariable **chain rule**.

Here is the step-by-step proof.

The Proof

1. Setting up two coordinate systems

Imagine you describe the points on your metal sheet using a standard Cartesian coordinate system, $x^\mu = (x^1, x^2)$.

Now, imagine a colleague decides to look at the same sheet using a different coordinate system, say polar coordinates or just a shifted/rotated grid, which we will call $\tilde{x}^\nu = (\tilde{x}^1, \tilde{x}^2)$.

The temperature at any physical point on the sheet is an intrinsic physical reality—it doesn't care what numbers you write on your grid. Therefore, f is a **scalar field**, meaning its value at a given point is invariant under a change of coordinates:

$$\tilde{f}(\tilde{x}) = f(x)$$

2. Taking the Gradient

In your coordinate system, the components of the gradient are given by the partial derivatives:

$$\nabla_\mu f = \frac{\partial f}{\partial x^\mu}$$

In your colleague's coordinate system, the components are:

$$\nabla_\nu \tilde{f} = \frac{\partial \tilde{f}}{\partial \tilde{x}^\nu}$$

3. Applying the Chain Rule

To see how your colleague's gradient components relate to yours, we apply the multi-variable chain rule to express the derivative with respect to $\tilde{x}^\nu$:

$$\frac{\partial \tilde{f}}{\partial \tilde{x}^\nu} = \frac{\partial f}{\partial x^\mu} \frac{\partial x^\mu}{\partial \tilde{x}^\nu}$$

(Note: We are using Einstein summation notation here, where the repeated index μ implies a sum over $\mu = 1, 2$.)

4. Analyzing the Transformation Law

Let's rewrite that result by substituting our gradient components:

$$\nabla_\nu \tilde{f} = \frac{\partial x^\mu}{\partial \tilde{x}^\nu} \nabla_\mu f$$

Look closely at the Jacobian matrix element $\frac{\partial x^\mu}{\partial \tilde{x}^\nu}$. Because the “new” coordinates $\tilde{x}$ are in the *denominator*, this is the exact mathematical definition of a **covariant transformation law**.

By definition, a geometric object whose components transform like:

$$V'_\nu = \frac{\partial x^\mu}{\partial x'^\nu} V_\mu$$

is a covariant vector. Therefore, the gradient ∇f is a covariant vector field.

Physical Intuition: Why “Co”-variant?

It helps to contrast this with regular positional vectors (like a displacement vector dx^μ), which are **contravariant**.

If you decide to shrink the scale of your grid lines (e.g., measuring in millimeters instead of centimeters), the *numerical values* of your coordinates change.

- **Contravariant (e.g., Distance):** The distance numbers get **larger** (1 cm becomes 10 mm). The components vary *contrary* to the change of scale.
- **Covariant (e.g., Gradient):** The temperature gradient measures *change per unit distance* ($^\circ\text{C}/\text{cm}$). If you switch to millimeters, the temperature change per millimeter gets **smaller** ($1^\circ\text{C}/\text{cm}$ becomes $0.1^\circ\text{C}/\text{mm}$). The components vary *together* with the scale change.

Because the gradient components live naturally as coefficients to displacements (forming the invariant scalar $df = \frac{\partial f}{\partial x^\mu} dx^\mu$), the gradient inherently belongs to the dual space, acting as a machine that eats contravariant vectors to output a scalar temperature change.